

The objective evolved from prediction alone to three explicit jobs

Legacy objective

$$L = L_{\text{JEPA}}$$

Centered and sharpened latent cross-entropy trained prediction.
Collapse was monitored, not actively regularized.

Current objective

$$L = L_{\text{JEPA}} + 0.05 L_{\text{VICReg}} + 0.25 L_{\text{group}}$$

JEPA: predict authorized hidden teacher features
VICReg: keep variation and reduce redundancy
Group: compact labels and penalize close centroids
Group is off during Stage 0

Feature std

0.414

evidence against total collapse

Normal anchor

0.594

substantial drift

Margin penalty

0.038

requested margin not fully met

Silhouette

0.009

weak canonical geometry

The group term reads condition labels. Only Stage 0 is label-free; Stages 1 through 4 are label-informed representation fine-tuning.